

Course Code	Course Title	L	T	P	C
MAMHA506	Artificial Intelligence for Mechatronics	3	1	0	4
Pre-requisite	NIL	Syllabus Version			
		1			
Course Objectives					
1. Introduce the principles and methodologies of Artificial Intelligence relevant to intelligent mechatronic systems.					
2. Develop the ability to apply AI techniques such as machine learning, neural networks and fuzzy systems in mechatronics applications.					
3. Provide practical exposure to AI-based design, modelling and control using modern tools and platforms.					
Course Outcomes					
1. Understand and explain the foundational concepts of AI and its applications in Mechatronic systems.					
2. Apply machine learning algorithms for different applications in intelligent mechatronic applications.					
3. Design and implement neural network and deep learning architectures.					
4. Develop fuzzy logic and hybrid intelligent systems for decision-making and control.					
5. Apply AI tools and techniques to design and evaluate intelligent mechatronic systems.					
Module:1	Introduction to AI and Data Science	8 hours			
Fundamentals of AI and its role in Mechatronics – Intelligent systems: Agents and environments – Introduction to Data Science: Data lifecycle, types of data – Data preprocessing, visualization and exploratory data analysis – Basics of statistics and probability in AI – Use of AI tools and platforms.					
Module:2	Machine Learning Techniques	9 hours			
Overview of Machine Learning in Mechatronics – Supervised Learning: Regression, Classification, Decision Trees, k-NN – Unsupervised Learning: Clustering (K-means, Hierarchical), Dimensionality Reduction – Reinforcement Learning: Markov Decision Process, Q-Learning – Model selection, training, validation, overfitting and bias-variance tradeoff.					
Module:3	Neural Networks and Deep Learning	9 hours			
Introduction to Artificial Neural Networks (ANN) – Multilayer Perceptron, Backpropagation algorithm – Deep Learning fundamentals: CNN, RNN, Autoencoders – Activation functions, loss functions and optimizers – Training strategies: Batch Normalization, Dropout, Early stopping.					
Module:4	Fuzzy Logic and Expert Systems	9 hours			

Fuzzy sets and membership functions – Linguistic variables, fuzzy rules and reasoning – Fuzzy Inference Systems: Mamdani and Sugeno models – Adaptive Neuro-Fuzzy Inference System (ANFIS) – Architecture and development of rule-based expert systems.		
Module:5	AI Applications in Mechatronics	8 hours
AI in robotics: path planning, object recognition and grasping – Predictive maintenance using AI – Intelligent control of actuators and drives – Smart sensors and AI-based signal processing – Human-machine interfaces using AI.		
Module:6	Contemporary Issues	2 hours
Total Lecture Hours:		45 hours
Tutorial Hours:		15 hours
Text Book(s)		
1. Russell, S., & Norvig, P, " Artificial Intelligence: A Modern Approach ", Pearson, 4 th Edition, 2020 2. Géron, A, " Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow ", O'Reilly Media, 3 rd Edition, 2022		
Reference Books		
1. Negnevitsky, M, " Artificial Intelligence: A Guide to Intelligent Systems ", Pearson, 4 th Edition, 2021 2. Goodfellow, I., Bengio, Y., & Courville, A, " Deep Learning ", MIT Press, 1 st Edition, 2021		
Mode of Evaluation :Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test		
Recommended by Board of Studies :		26-05-2025
Approved by Academic Council : No. 78		12-06-2025